Further Extensions of Haglund-Remmel-Wilson Identity

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Abstract. MacMahon's equidistribution theorem states that the permutation statistics inversion number and major index are equidistributed. In 2015, Remmel and Wilson proved a conjectured identity of Haglund which is an extension of MacMahon's equidistribution theorem to ordered set partitions. Recently, Liu extended this identity to k-Stirling permutations and posed a conjecture concerning an ascent analog of his extension. In this paper, we shall present a combinatorial proof of this conjecture. Furthermore, we derive an analogous result for another maj−like statistic introduced by Liu.

Keywords: Stirling permutation, inversion number, major index, MacMahon's equidistribution theorem

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1 Introduction

Let $\mathcal{S}_n$ denote the set of permutations of $[n] = \{1, 2, \ldots, n\}$. Given $\pi = \pi_1\pi_2\cdots\pi_n \in \mathcal{S}_n$, a pair (i, j) is called an *inversion* of π if $i < j$ and $\pi_i > \pi_j$. Let $\mathrm{Inv}(\pi)$ denote the set of inversions of π and let $\mathrm{inv}(\pi)$ denote the number of inversions of π. Let

$$\mathrm{inv}^{\square,\mathrm{j}}(\pi) = |\{i \mid (i, j) \in \mathrm{Inv}(\pi)\}|$$

be the number of inversions of π that end at position j. Define the *descent set* and the *ascent set* of π to be

$$\mathrm{Des}(\pi) = \{i \mid \pi_i > \pi_{i+1}, i \in [n-1]\},$$

and

$$\mathrm{Asc}(\pi) = \{i \mid \pi_i < \pi_{i+1}, i \in [n-1]\},$$

respectively. Let $\mathrm{des}(\pi) = |\mathrm{Des}(\pi)|$ and $\mathrm{asc}(\pi) = |\mathrm{Asc}(\pi)|$. The *major index* of π, denoted by $\mathrm{maj}(\pi)$, is defined to be

$$\mathrm{maj}(\pi) = \sum_{i \in \mathrm{Des}(\pi)} i.$$

The *comajor index* of π, denoted by $\mathrm{comaj}(\pi)$, is defined to be

$$\mathrm{comaj}(\pi) = \sum_{i \in \mathrm{Asc}(\pi)} i.$$

MacMahon's equidistribution theorem [11] states that

$$\sum_{\pi \in \mathcal{S}_n} q^{\mathrm{inv}(\pi)} = \sum_{\pi \in \mathcal{S}_n} q^{\mathrm{maj}(\pi)}. \tag{1.1}$$

The first bijective proof of (1.1) was found by Foata [6] in 1968. Carlitz gave another combinatorial proof in [4].

By using the complement bijection, we have

$$\sum_{\pi \in \mathcal{S}_n} q^{\mathrm{inv}(\pi)} = \sum_{\pi \in \mathcal{S}_n} q^{\mathrm{maj}(\pi)} = \sum_{\pi \in \mathcal{S}_n} q^{\mathrm{comaj}(\pi)}. \tag{1.2}$$

In [14], Remmel and Wilson confirmed the following conjectured identity posed by Haglund:

$$\sum_{\pi \in \mathcal{S}_n} q^{\mathrm{inv}(\pi)} \prod_{j \in \mathrm{Des}(\pi)} \left(1 + \frac{z}{q^{1+\mathrm{inv}^{\square,\mathrm{j}}(\pi)}}\right) = \sum_{\pi \in \mathcal{S}_n} q^{\mathrm{maj}(\pi)} \prod_{j=1}^{\mathrm{des}(\pi)} \left(1 + \frac{z}{q^j}\right). \tag{1.3}$$

(1.3) is called Haglund-Remmel-Wilson identity in [10]. Specializing $z = 0$ in (1.3) recovers (1.1), so (1.3) can be viewed as an extension of MacMahon's equidistribution theorem. Remmel and Wilson [14] also proved the following ascent analogue of (1.3):

$$\sum_{\pi \in \mathcal{S}_n} q^{\mathrm{inv}(\pi)} \prod_{j \in \mathrm{Asc}(\pi)} \left(1 + \frac{z}{q^{\mathrm{inv}^{\square,j+1}(\pi)}}\right) = \sum_{\pi \in \mathcal{S}_n} q^{\mathrm{comaj}(\pi)} \prod_{j=1}^{\mathrm{asc}(\pi)} \left(1 + \frac{z}{q^j}\right). \tag{1.4}$$

Let $\mathcal{M} = \{1^{k_1}, 2^{k_2}, \ldots, n^{k_n}\}$ be a multiset where k_i is the number of occurrences of i in $\mathcal{M}$ and $k_i \geq 1$. A permutation π on a multiset $\mathcal{M}$ is said to be a *Stirling permutation* if $\pi_i = \pi_k$ implies that $\pi_j \geq \pi_i$ for all $i < j < k$. Stirling permutations were originally introduced by Gessel and Stanley [7] in the case of the multiset $\{1^2, 2^2, \ldots, n^2\}$. A Stirling permutation π on $\{1^k, 2^k, \ldots, n^k\}$ is said to be a *k-Stirling permutation* of order n. Let $\mathcal{Q}_n(k)$ denote the set of k-Stirling permutations of order n. There is extensive literature on Stirling permutations and k-Stirling permutations, see e.g. [1, 2, 3, 5, 8, 12, 13].

In order to find an analogue of (1.2) for k-Stirling permutations, Liu [9] introduced the following inv– and maj–like statistics for k-Stirling permutations. Given $\pi = \pi_1\pi_2\cdots\pi_{kn} \in \mathcal{Q}_n(k)$, the pair (i,j) is said to be a *last inversion* of π if the pair (i,j) is an inversion of π and π_i is the last occurrence of that letter in π. Let $\mathrm{Linv}(\pi)$ denote the set of last inversions of π and let $\mathrm{linv}(\pi)$ denote the number of the last inversions of π. An index i, $1 \le i \le k(n-1)$, is said to be a *longest plateau-descent* if

$$\pi_i = \pi_{i+1} = \cdots = \pi_{i+k-1} > \pi_{i+k}.$$

Let $\mathrm{Lpd}(\pi)$ denote the set of the longest plateau-descents of π and let $\mathrm{lpd}(\pi) = |\mathrm{Lpd}(\pi)|$. An index i, $1 \le i \le k(n-1)$, is said to be a *longest ascent-plateau* if

$$\pi_i < \pi_{i+1} = \pi_{i+2} = \cdots = \pi_{i+k}.$$

Let $\mathrm{Lap}(\pi)$ denote the set of the longest ascent-plateaux of π and let $\mathrm{lap}(\pi) = |\mathrm{Lap}(\pi)|$. Define the longest major index of π, denoted by lmaj, to be

$$\mathrm{lmaj}(\pi) = \sum_{i \in \mathrm{Lpd}(\pi)} i.$$

Similarly, the longest comajor index of π, denoted by lcomaj, is defined to be

$$\mathrm{lcomaj}(\pi) = \sum_{i \in \mathrm{Lap}(\pi)} i.$$

For example, if $\pi = 124445552233311 \in \mathcal{Q}_5(3)$, then

$$\mathrm{Lpd}(\pi) = \{6, 11\},\ \mathrm{Lap}(\pi) = \{2, 5, 10\},$$

and

$$\mathrm{linv}(\pi) = 18,\ \ \mathrm{lmaj}(\pi) = 17,\ \ \mathrm{lcomaj}(\pi) = 17.$$

Restricted to ordinary permutations, Linv (resp. linv, Lpd, Lap, lpd, lap, lmaj, lcomaj) coincides with Inv (resp. inv, Des, Asc, des, asc, maj, comaj).

In [9], Liu derived that

$$\sum_{\pi \in \mathcal{Q}_n(k)} q^{\mathrm{linv}(\pi)} = \sum_{\pi \in \mathcal{Q}_n(k)} q^{\mathrm{lmaj}(\pi)} = \sum_{\pi \in \mathcal{Q}_n(k)} q^{\mathrm{lcomaj}(\pi)}, \tag{1.5}$$

which is an extension of (1.2) to k-Stirling permutations.

In analogy to ordinary permutations, given a k-Stirling permutation π, let

$$\mathrm{linv}^{\square,\mathrm{j}}(\pi) = |\{i \mid (i,j) \in \mathrm{Linv}(\pi)\}|$$

denote the number of last inversions of π which end at position j. Very recently, Liu [10] obtained the following extension of Haglund-Remmel-Wilson identity to k-Stirling permutations:

$$\sum_{\pi\in\mathcal{Q}_n(k)} q^{\mathrm{linv}(\pi)} \prod_{j\in \mathrm{Lpd}(\pi)} (1+\frac{z}{q^{1+k\cdot \mathrm{linv}^{\square,j}(\pi)}}) = \sum_{\pi\in\mathcal{Q}_n(k)} q^{\mathrm{lmaj}(\pi)} \prod_{j=1}^{\mathrm{lpd}(\pi)} (1+\frac{z}{q^{k(j-1)+1}}). \quad (1.6)$$

Specializing $k=1$ in (1.6) recovers Haglund-Remmel-Wilson identity (1.3).

Motivated by finding an extension of (1.4) to k-Stirling permutations, Liu [10] posed the following conjecture.

Conjecture 1.1 (See [10]) *For all $n,k\geq 1$, we have*

$$\sum_{\pi\in\mathcal{Q}_n(k)} q^{\mathrm{linv}(\pi)} \prod_{j\in \mathrm{Lap}(\pi)} (1+\frac{z}{q^{k\cdot \mathrm{linv}^{\square,j+1}(\pi)}}) = \sum_{\pi\in\mathcal{Q}_n(k)} q^{\mathrm{lcomaj}(\pi)} \prod_{j=1}^{\mathrm{lap}(\pi)} (1+\frac{z}{q^{k(j-1)+1}}). \quad (1.7)$$

Notice that specializing $k=1$ in (1.7) recovers (1.4).

In [9], Liu also introduced the following complicated maj-like statistic for k-Stirling permutations. Given $\pi=\pi_1\pi_2\cdots\pi_{kn}\in\mathcal{Q}_n(k)$, an index i, $1\leq i\leq k(n-1)$, is said to be a *plateau-peak* if

$$\pi_i<\pi_{i+1}=\pi_{i+2}=\cdots=\pi_{i+k}>\pi_{i+k+1}$$

with the assumption $\pi_{kn+1}=0$. Let $\mathrm{Pp}(\pi)$ denote the set of plateau-peaks of π and let $\mathrm{pp}(\pi)=|\mathrm{Pp}(\pi)|$. For $1\leq i\leq n$, let $\pi_{(i)}$ denote the subsequence of π consisting of the letters that lie weakly to the right of the first occurrence of letter i and that are not greater than i. Define the pp-maj of π to be

$$\text{pp-maj}(\pi)=\sum_{i\in\mathrm{Pp}(\pi)} i+\sum_{i=1}^{n}\mathrm{pp}(\pi_{(i)}).$$

For example, if $\pi=124445552233311\in\mathcal{Q}_5(3)$, then we have

$$\mathrm{Pp}(\pi)=\{5,10\},$$

$$\begin{array}{ll}
\pi_{(1)}=111 & \mathrm{pp}(\pi_{(1)})=0,\\
\pi_{(2)}=22211 & \mathrm{pp}(\pi_{(2)})=0,\\
\pi_{(3)}=33311 & \mathrm{pp}(\pi_{(3)})=0,\\
\pi_{(4)}=4442233311 & \mathrm{pp}(\pi_{(4)})=1,\\
\pi_{(5)}=5552233311 & \mathrm{pp}(\pi_{(5)})=1,
\end{array}$$

and

$$\begin{aligned}\text{pp-maj}(\pi) &= \sum_{i\in \mathrm{Pp}(\pi)} i + \sum_{i=1}^{n} \mathrm{pp}(\pi_{(i)}) \\ &= 15+2 \\ &= 17.\end{aligned}$$

Liu [9] proved that

$$\sum_{\pi\in\mathcal{Q}_n(k)} q^{\mathrm{linv}(\pi)} = \sum_{\pi\in\mathcal{Q}_n(k)} q^{\mathrm{pp-maj}(\pi)}. \tag{1.8}$$

Our second result is concerned with the following analogous result of Conjecture 1.1.

Theorem 1.2 *For all $n, k \geq 1$, we have*

$$\sum_{\pi\in\mathcal{Q}_n(k)} q^{\mathrm{linv}(\pi)} \prod_{j\in \mathrm{Pp}(\pi)} \left(1+\frac{z}{q^{(k+1)\cdot \mathrm{linv}^{\square,j+1}(\pi)}}\right) = \sum_{\pi\in\mathcal{Q}_n(k)} q^{\mathrm{pp-maj}(\pi)} \prod_{j=1}^{\mathrm{pp}(\pi)} \left(1+\frac{z}{q^{(k+1)(j-1)+1}}\right). \tag{1.9}$$

As specializing $z=0$ recovers (1.8), (1.9) can be viewed as a generalization of (1.8). In this paper, we shall present combinatorial proofs of Conjecture 1.1 and Theorem 1.2 in the spirit of Remmel and Wilson's combinatorial proof of (1.3) and Liu's combinatorial proof of (1.6).

2 Proof of Conjecture 1.1

Given a permutation $\pi \in \mathcal{Q}_n(k)$ and a set $S \subseteq \mathrm{Lap}(\pi)$, the pair (π, S) is said to be a *longest ascent-plateau starred k-Stirling permutation* if for each $i \in S$, we put a star at each position after π_j for all $j \in \{i, i+1, \ldots, i+k-1\}$. For example, for $\pi = 124445552233311$ and $S = \{2, 10\}$, we have $(\pi, S) = 12_*4_*4_*455522_*3_*3_*311$.

Define

$$\mathcal{Q}_n^{\mathrm{Lap}}(k) = \{(\pi, S) \mid \pi \in \mathcal{Q}_n(k), S \subseteq \mathrm{Lap}(\pi)\}$$

and

$$\mathcal{Q}_n^{\mathrm{Lap}}(k, s) = \{(\pi, S) \mid (\pi, S) \in \mathcal{Q}_n^{\mathrm{Lap}}(k), |S| = s\}.$$

For any permutation $(\pi, S) \in \mathcal{Q}_n^{\mathrm{Lap}}(k)$, a letter π_i is said to be a *starred* (resp. *unstarred*) letter if the position before π_i is starred (resp. unstarred). Define the *last inversion* of (π, S), denoted by $\mathrm{linv}(\pi, \mathrm{S})$, to be

$$\mathrm{linv}(\pi, \mathrm{S}) = \mathrm{linv}(\pi) - \sum_{j\in S} k \cdot \mathrm{linv}^{\square,\mathrm{j}+1}(\pi).$$

It is easy to check that $\mathrm{linv}(\pi, \mathrm{S})$ is the number of last inversions of π which end at unstarred letters for any $(\pi, S) \in \mathcal{Q}_n^{\mathrm{Lap}}(k)$. For example, if let $(\pi, S) = 12_*4_*4_*455522_*3_*3_*311 \in \mathcal{Q}_5^{\mathrm{Lap}}(3)$, then

$$\mathrm{Lap}(\pi) = \{2, 5, 10\},\ S = \{2, 10\}$$

and

$$\begin{aligned} \mathrm{linv}(\pi, \mathrm{S}) &= \mathrm{linv}(\pi) - \sum_{j \in S} k \cdot \mathrm{linv}^{\square, \mathrm{j}+1}(\pi) \\ &= 18 - 3 \cdot \mathrm{linv}^{\square,3}(\pi) - 3 \cdot \mathrm{linv}^{\square,11}(\pi) \\ &= 18 - 0 - 3 \times 2 \\ &= 12. \end{aligned}$$

Define

$$Q_{n,k,s}^{\mathrm{Lap}}(q) = \sum_{(\pi,S) \in \mathcal{Q}_n^{\mathrm{Lap}}(k,s)} q^{\mathrm{linv}(\pi, \mathrm{S})}.$$

We have the following lemma for the left-hand side of Conjecture 1.1.

Lemma 2.1 *We have*

$$\sum_{\pi \in \mathcal{Q}_n(k)} q^{\mathrm{linv}(\pi)} \prod_{j \in \mathrm{Lap}(\pi)} \left(1 + \frac{z}{q^{k \cdot \mathrm{linv}^{\square, j+1}(\pi)}}\right) = \sum_{s \geq 0} Q_{n,k,s}^{\mathrm{Lap}}(q) z^s. \tag{2.1}$$

Proof. We have

$$\begin{aligned} \sum_{\pi \in \mathcal{Q}_n(k)} q^{\mathrm{linv}(\pi)} \prod_{j \in \mathrm{Lap}(\pi)} \left(1 + \frac{z}{q^{k \cdot \mathrm{linv}^{\square, j+1}(\pi)}}\right) &= \sum_{\pi \in \mathcal{Q}_n(k)} \sum_{S \subseteq \mathrm{Lap}(\pi)} q^{\mathrm{linv}(\pi)} \prod_{j \in S} z q^{-k \cdot \mathrm{linv}^{\square, j+1}(\pi)} \\ &= \sum_{s \geq 0} \sum_{(\pi,S) \in \mathcal{Q}_n^{\mathrm{Lap}}(k,s)} q^{\mathrm{linv}(\pi, \mathrm{S})} z^s \\ &= \sum_{s \geq 0} Q_{n,k,s}^{\mathrm{Lap}}(q) z^s, \end{aligned}$$

as desired. ∎

We shall derive a recurrence relation for $Q_{n,k,s}^{\mathrm{Lap}}(q)$. To this end, first we give a linv-labeling for permutations in $\mathcal{Q}_n^{\mathrm{Lap}}(k)$. Given $(\pi, S) \in \mathcal{Q}_n^{\mathrm{Lap}}(k, s)$, the linv-labeling of (π, S) is obtained by labeling the unstarred positions of π by $0, 1, \ldots, kn - ks$ from right to left. For example, for $(\pi, S) = 12_*4_*4_*455522_*3_*3_*311$, the linv-labeling of (π, S) is

$$_9 1_8 2_* 4_* 4_* 4_7 5_6 5_5 5_4 2_3 2_* 3_* 3_* 3_2 1_1 1_0,$$

where the labels of the positions are written as subscripts.

Denote by $\bar{\mathcal{Q}}_n^{\mathrm{Lap}}(k, s)$ (resp. $\hat{\mathcal{Q}}_n^{\mathrm{Lap}}(k, s)$) the set of starred permutations $(\pi, S) \in \mathcal{Q}_n^{\mathrm{Lap}}(k, s)$ such that all the $n's$ are unstarred (resp. starred) in (π, S). Clearly, the

sets $\bar{\mathcal{Q}}_n^{\mathrm{Lap}}(k,s)$ and $\hat{\mathcal{Q}}_n^{\mathrm{Lap}}(k,s)$ are disjoint and the union of these two sets is exactly $\mathcal{Q}_n^{\mathrm{Lap}}(k,s)$. Notice that a starred permutation in $\bar{\mathcal{Q}}_n^{\mathrm{Lap}}(k,s)$ can be generated by a starred permutation π in $\mathcal{Q}_{n-1}^{\mathrm{Lap}}(k,s)$ by inserting k copies of n at some unstarred position of π. While a starred permutation in $\hat{\mathcal{Q}}_n^{\mathrm{Lap}}(k,s)$ can be generated by a starred permutation π in $\mathcal{Q}_{n-1}^{\mathrm{Lap}}(k,s-1)$ by inserting k copies of n at some unstarred position of π (except the position before π) and labeling the position before each n by a star. We shall describe the generative process of starred permutations in $\mathcal{Q}_n^{\mathrm{Lap}}(k,s)$ by building two bijections.

We first construct a map

$$\phi_{n,k,s} : \{0,1,\ldots,kn-ks-k\} \times \mathcal{Q}_{n-1}^{\mathrm{Lap}}(k,s) \longrightarrow \bar{\mathcal{Q}}_n^{\mathrm{Lap}}(k,s)$$

by mapping the pair $(i,(\pi,S))$ to the starred permutation by inserting k copies of n at the position in (π,S) which is labeled by i in the linv-labeling of (π,S). For example, we have

$$\phi_{6,3,2}(5,(\pi,S)) = 12_*4_*4_*455666522_*3_*3_*311,$$

where $(\pi,S) = 12_*4_*4_*455522_*3_*3_*311$.

Similarly, we define a map

$$\phi^*_{n,k,s} : \{0,1,\ldots,kn-ks-1\} \times \mathcal{Q}_{n-1}^{\mathrm{Lap}}(k,s-1) \longrightarrow \hat{\mathcal{Q}}_n^{\mathrm{Lap}}(k,s)$$

by mapping the pair $(i,(\pi,S))$ to the starred permutation by inserting k copies of n at the position in (π,S) which is labeled by i in the linv-labeling of (π,S) and labeling the position before each n by a star. Here we should note that we cannot insert k copies of n at the start of π by the definition of the starred permutations. For example, we have

$$\phi^*_{6,3,3}(5,(\pi,S)) = 12_*4_*4_*455_*6_*6_*6522_*3_*3_*311,$$

where $(\pi,S) = 12_*4_*4_*455522_*3_*3_*311$. One can easily check that both $\phi_{n,k,s}$ and $\phi^*_{n,k,s}$ are bijections. The next lemma establishes the relationship between these labels and the insertion maps.

Lemma 2.2 *We have*

- *for* $(\pi,S) \in \mathcal{Q}_{n-1}^{\mathrm{Lap}}(k,s)$ *and* $0 \le i \le kn-ks-k$,

$$\mathrm{linv}(\phi_{n,k,s}(i,(\pi,S))) = \mathrm{linv}(\pi,S) + i;$$

- *for* $(\pi,S) \in \mathcal{Q}_{n-1}^{\mathrm{Lap}}(k,s-1)$ *and* $0 \le i \le kn-ks-1$,

$$\mathrm{linv}(\phi^*_{n,k,s}(i,(\pi,S))) = \mathrm{linv}(\pi,S) + i.$$

Proof. According to the construction of the map $\phi_{n,k,s}$, inserting k copies of n at the position labeled by i in the linv-labeling creates i new last inversions (between the last appearance of n and each unstarred letter to its right) and does not affect the last inversions to the left of the leftmost n. The same is true for the map $\phi^*_{n,k,s}$. This completes the proof. ∎

For a nonnegative integer n, let $[n]_q = 1$ when $n = 0$ and $[n]_q = 1 + q + \cdots + q^{n-1}$ otherwise. Then we have the following recurrence relation for $Q^{\mathrm{Lap}}_{n,k,s}(q)$.

Lemma 2.3 *For $n \geq 2$, the polynomial $Q^{\mathrm{Lap}}_{n,k,s}(q)$ equals 0 when $s < 0$ or $s > n-1$ and*

$$Q^{\mathrm{Lap}}_{n,k,s}(q) = [kn - ks - k + 1]_q Q^{\mathrm{Lap}}_{n-1,k,s}(q) + [kn - ks]_q Q^{\mathrm{Lap}}_{n-1,k,s-1}(q)$$

otherwise.

Proof. It follows from Lemma 2.2 and the properties of our bijections $\phi_{n,k,s}$ and $\phi^*_{n,k,s}$ that

$$\begin{aligned}
Q^{\mathrm{Lap}}_{n,k,s}(q) &= \sum_{(\pi,S)\in\mathcal{Q}^{\mathrm{Lap}}_n(k,s)} q^{\mathrm{linv}(\pi,S)} \\
&= \sum_{(\pi,S)\in\bar{\mathcal{Q}}^{\mathrm{Lap}}_n(k,s)} q^{\mathrm{linv}(\pi,S)} + \sum_{(\pi,S)\in\hat{\mathcal{Q}}^{\mathrm{Lap}}_n(k,s)} q^{\mathrm{linv}(\pi,S)} \\
&= \sum_{i=0}^{kn-ks-k} \sum_{(\pi,S)\in\mathcal{Q}^{\mathrm{Lap}}_{n-1}(k,s)} q^{i+\mathrm{linv}(\pi,S)} + \sum_{i=0}^{kn-ks-1} \sum_{(\pi,S)\in\mathcal{Q}^{\mathrm{Lap}}_{n-1}(k,s-1)} q^{i+\mathrm{linv}(\pi,S)} \\
&= [kn - ks - k + 1]_q Q^{\mathrm{Lap}}_{n-1,k,s}(q) + [kn - ks]_q Q^{\mathrm{Lap}}_{n-1,k,s-1}(q),
\end{aligned}$$

as desired. ∎

Next, we consider the right-hand side of Conjecture 1.1. For $\pi \in \mathcal{Q}_n(k)$, if $i \in \mathrm{Lap}(\pi)$, the set $[i, i+k) = \{i, i+1, \ldots, i+k-1\}$ is called a *longest ascent-plateau block* of π. Let $\overline{\mathrm{Lap}}(\pi) = \bigcup_{i\in\mathrm{Lap}(\pi)} [i, i+k)$. For $(\pi, S) \in \mathcal{Q}^{\mathrm{Lap}}_n(k)$, if $i \in S$, we say that the longest ascent-plateau block $[i, i+k)$ is starred. Note that all the positions in a starred longest ascent-plateau block are starred.

In order to give a combinatorial interpretation of the product $\prod_{j=1}^{\mathrm{lap}(\pi)} (1 + \frac{z}{q^{k(j-1)+1}})$ in the right-hand side of Conjecture 1.1, we think the index j in the product as the referring to the longest ascent-plateau block (counting from right to left). The choice of 1 from the factor $1 + \frac{z}{q^{k(j-1)+1}}$ is considered as leaving the j-th longest ascent-plateau block unstarred, while the choice of $\frac{z}{q^{k(j-1)+1}}$ from the factor $1 + \frac{z}{q^{k(j-1)+1}}$ is considered as starring the j-th longest ascent-plateau block. With this in mind, we

define the *longest comajor index* of a longest ascent-plateau starred permutation $(\pi, S) \in \mathcal{Q}_n^{\mathrm{Lap}}(k)$, denoted by $\mathrm{lcomaj}(\pi, \mathrm{S})$, to be

$$\mathrm{lcomaj}(\pi, \mathrm{S}) = \mathrm{lcomaj}(\pi) - |S| - \sum_{j \in S} k \cdot |\mathrm{Lap}(\pi) \cap \{j+k, \ldots, kn\}|.$$

Clearly, the term $|\mathrm{Lap}(\pi) \cap \{j+k, \ldots, kn\}|$ can be interpreted as the number of longest ascent-plateaux to the right of the starred longest ascent-plateau block $[j, j+k)$ for any $j \in S$. As there are k-starred positions in the starred longest ascent-plateau block $[j, j+k)$, the sum $\sum_{j \in S} k \cdot |\mathrm{Lap}(\pi) \cap \{j+k, \ldots, kn\}|$ actually counts the sum of the number of starred positions to the left of π_j for any $j \in \mathrm{Lap}(\pi)$. Hence, we have

$$\mathrm{lcomaj}(\pi, \mathrm{S}) = \mathrm{lcomaj}(\pi) - |S| - \sum_{j \in \mathrm{Lap}(\pi)} k \cdot |S \cap \{1, 2, \ldots, j-1\}|.$$

For example, if $(\pi, S) = 12_*4_*4_*455522_*3_*3_*311$, then

$$\mathrm{Lap}(\pi) = \{2, 5, 10\},\ S = \{2, 10\}$$

and

$$\begin{aligned} \mathrm{lcomaj}(\pi, \mathrm{S}) &= \mathrm{lcomaj}(\pi) - |S| - \sum_{j \in \mathrm{Lap}(\pi)} k \cdot |S \cap \{1, 2, \ldots, j-1\}| \\ &= 17 - 2 - 0 - 3 - 3 \\ &= 9. \end{aligned}$$

Define

$$\tilde{Q}_{n,k,s}^{\mathrm{Lap}}(q) = \sum_{(\pi, S) \in \mathcal{Q}_n^{\mathrm{Lap}}(k,s)} q^{\mathrm{lcomaj}(\pi, \mathrm{S})}.$$

Through the above analysis, we have the following lemma for the right-hand side of Conjecture 1.1.

Lemma 2.4 *We have*

$$\sum_{\pi \in \mathcal{Q}_n(k)} q^{\mathrm{lcomaj}(\pi)} \prod_{j=1}^{\mathrm{lap}(\pi)} \left(1 + \frac{z}{q^{k(j-1)+1}}\right) = \sum_{s \geq 0} \tilde{Q}_{n,k,s}^{\mathrm{Lap}}(q) z^s.$$

Proof. We have

$$\begin{aligned} \sum_{\pi \in \mathcal{Q}_n(k)} q^{\mathrm{lcomaj}(\pi)} \prod_{j=1}^{\mathrm{lap}(\pi)} \left(1 + \frac{z}{q^{k(j-1)+1}}\right) &= \sum_{\pi \in \mathcal{Q}_n(k)} q^{\mathrm{lcomaj}(\pi)} \Big(1 + \sum_{s \geq 1} \sum_{1 \leq j_1 < j_2 < \cdots < j_s \leq \mathrm{lap}(\pi)} \Big(\prod_{j \in \{j_1, j_2, \ldots, j_s\}} \frac{z}{q^{k(j-1)+1}} \Big) \Big) \\ &= \sum_{s \geq 0} \sum_{(\pi, S) \in \mathcal{Q}_n^{\mathrm{Lap}}(k,s)} q^{\mathrm{lcomaj}(\pi, \mathrm{S})} z^s \\ &= \sum_{s \geq 0} \tilde{Q}_{n,k,s}^{\mathrm{Lap}}(q) z^s, \end{aligned}$$

as desired. ∎

In the rest of this section, we will devote to deriving a recurrence relation for $\tilde{Q}^{\mathrm{Lap}}_{n,k,s}(q)$, which turns out to be the same as the one for $Q^{\mathrm{Lap}}_{n,k,s}(q)$. This allows us to prove Conjecture 1.1 immediately.

Now we give a lcomaj-labeling for permutations in $\mathcal{Q}^{\mathrm{Lap}}_n(k)$. For $(\pi, S) \in \mathcal{Q}^{\mathrm{Lap}}_n(k)$ with $|S| = s$ and $\mathrm{lap}(\pi) = r$, the lcomaj-labeling of (π, S) is obtained as follows.

- Label the positions in the rightmost unstarred longest ascent-plateau block from left to right by $0, 1, \ldots, k-1$, label the positions in the second to the rightmost unstarred longest ascent-plateau block from left to right by $k, k+1, \ldots, 2k-1$, and so on.
- Label the remaining unstarred positions from left to right by $kr-ks, kr-ks+1, kr-ks+2, \ldots, kn-ks$.

Since the leftmost position is always unstarred, the label of this position is $kr-ks$.

For example, given $(\pi, S) = 16662_*4_*4_*455522_*3_*3_*311$, the lcomaj-labeling of (π, S) is

$$_6 1_3 6_4 6_5 6_7 2_* 4_* 4_* 4_0 5_1 5_2 5_8 2_9 2_* 3_* 3_* 3_{10} 1_{11} 1_{12},$$

where the labels of the positions are written as subscripts.

Let $\mathcal{X}\mathcal{Q}^{\mathrm{Lap}}_n(k,s)$ denote the set of starred permutations $(\pi, S) \in \mathcal{Q}^{\mathrm{Lap}}_n(k,s)$ whose rightmost longest ascent-plateau block (if any) is unstarred. Let $\mathcal{Y}\mathcal{Q}^{\mathrm{Lap}}_n(k,s)$ denote the set of starred permutations $(\sigma, S) \in \mathcal{Q}^{\mathrm{Lap}}_n(k,s)$ satisfying that

(i) $[p, p+k)$ is the rightmost longest ascent-plateau block of σ where $\sigma_{p+i} = n$ for all $1 \le i \le k$;

(ii) the rightmost longest ascent-plateau block $[p, p+k)$ is starred;

(iii) $\mathrm{Lap}(\pi) = \mathrm{Lap}(\sigma)$ where π is obtained from σ by removing all the $n's$.

Define

$$\mathcal{Z}\mathcal{Q}^{\mathrm{Lap}}_n(k,s) = \mathcal{Q}^{\mathrm{Lap}}_n(k,s) - \mathcal{X}\mathcal{Q}^{\mathrm{Lap}}_n(k,s) - \mathcal{Y}\mathcal{Q}^{\mathrm{Lap}}_n(k,s).$$

Lemma 2.5 *For $n \ge 2$ and $s \ge 0$, there is a bijection*

$$\delta_{n,k,s} : \{0, 1, \ldots, kn-ks-k\} \times \mathcal{Q}^{\mathrm{Lap}}_{n-1}(k,s) \longrightarrow \mathcal{X}\mathcal{Q}^{\mathrm{Lap}}_n(k,s) \bigcup \mathcal{Y}\mathcal{Q}^{\mathrm{Lap}}_n(k,s)$$

such that $\mathrm{lcomaj}(\delta_{n,k,s}(i, (\pi, S))) = \mathrm{lcomaj}(\pi, S) + i$ *for any* $(\pi, S) \in \mathcal{Q}^{\mathrm{Lap}}_{n-1}(k,s)$ *and* $0 \le i \le kn-ks-k$.

Proof. First, we give a description of the map $\delta_{n,k,s}$. Let $(i,(\pi,S)) \in \{0,1,\ldots,kn-ks-k\} \times \mathcal{Q}_{n-1}^{\mathrm{Lap}}(k,s)$ with $\mathrm{lap}(\pi) = r$. Now we shall demonstrate how to construct $\delta_{n,k,s}(i,(\pi,S))$ and show that $\mathrm{lcomaj}(\delta_{n,k,s}(i,(\pi,S))) = \mathrm{lcomaj}(\pi,S) + i$ by distinguishing the following two cases.

Case (i): $i = kr - ks$ with $r \geq 1$ and the leftmost longest ascent-plateau block of (π,S) is starred. We construct $\delta_{n,k,s}(i,(\pi,S))$ by the following procedure.

1. Remove the stars in the leftmost longest ascent-plateau block;
2. Move all the stars in each longest ascent-plateau block one (longest ascent-plateau) block to its left;
3. Let (π,T) be the resulting starred permutation and let $[p,p+k)$ be the rightmost longest ascent-plateau block in (π,T);
4. Insert k copies of n at the position immediately following π_p and label the position before each n by a star.

The second step is well-defined since the leftmost longest ascent-plateau block will be unstarred after the first step. Notice that the rightmost longest ascent-plateau block $[p,p+k)$ is unstarred after the second step. This ensures that the fourth step is well-defined. From the construction of $\delta_{n,k,s}(i,(\pi,S))$, one can easily verify that $\delta_{n,k,s}(i,(\pi,S))$ verifies properties (i)-(iii) of $\mathcal{YQ}_n^{\mathrm{Lap}}(k,s)$. Hence, we have $\delta_{n,k,s}(i,(\pi,S)) \in \mathcal{YQ}_n^{\mathrm{Lap}}(k,s)$.

For example, let $(\pi,S) = 12_*4_*4_*455522_*3_*3_*311 \in \mathcal{Q}_5^{\mathrm{Lap}}(3,2)$ and $i = 3$. Clearly, the lcomaj-labeling of (π,S) is given by

$$ {}_3 1_4 2_* 4_* 4_* 4_0 5_1 5_2 5_5 2_6 2_* 3_* 3_* 3_7 1_8 1_9. $$

Removing the stars at the positions in the leftmost longest ascent-plateau block, we obtain $1244455522_*3_*3_*311$. Then, we get

$$ (\pi,T) = 12444_*5_*5_*52233311 $$

by moving all the stars in each longest ascent-plateau block one (longest ascent-plateau) block to its left. Finally, we obtain

$$ \delta_{6,3,2}(3,(\pi,S)) = 12444_*5_*5_*522_*6_*6_*633311 $$

by inserting three copies of 6 at the position immediately following $\pi_{10} = 2$ and labeling the position before each 6 by a star.

Recall that

$$\mathrm{lcomaj}(\pi, \mathrm{S}) = \mathrm{lcomaj}(\pi) - |S| - \sum_{j \in \mathrm{Lap}(\pi)} k \cdot |S \cap \{1, 2, \ldots, j-1\}|.$$

Removing the stars in the leftmost longest ascent-plateau block decreases $|S|$ by 1 and decreases the sum $\sum_{j \in \mathrm{Lap}(\pi)} k \cdot |S \cap \{1, 2, \ldots, j-1\}|$ by $k(r-1)$. Observe that moving all the stars in a longest ascent-plateau block one (longest ascent-plateau) block to its left increases the sum $\sum_{j \in \mathrm{Lap}(\pi)} k \cdot |S \cap \{1, 2, \ldots, j-1\}|$ by k. Since we are moving all the stars in the $s-1$ longest ascent-plateau blocks, the sum $\sum_{j \in \mathrm{Lap}(\pi)} k \cdot |S \cap \{1, 2, \ldots, j-1\}|$ will be increased by $k(s-1)$. Hence, we have

$$|T| + \sum_{j \in \mathrm{Lap}(\pi)} k \cdot |T \cap \{1, 2, \ldots, j-1\}| = k(s-1) - k(r-1) - 1 + |S| + \sum_{j \in \mathrm{Lap}(\pi)} k \cdot |S \cap \{1, 2, \ldots, j-1\}|.$$

Let $(\sigma, S') = \delta_{n,k,s}(i, (\pi, S))$. Observe that (σ, S') is obtained from (π, T) by inserting k copies of n at the position immediately following π_p and labeling the position before each n by a star. Clearly, this process does not affect the sum $\sum_{j \in \mathrm{Lap}(\pi)} k \cdot |T \cap \{1, 2, \ldots, j-1\}|$ as $[p, p+k)$ is the rightmost longest ascent-plateau block of π. Moreover, we have $S' = T \cup \{p\}$ and $\mathrm{Lap}(\pi) = \mathrm{Lap}(\sigma)$. Hence, we have

$$\begin{aligned}
\mathrm{lcomaj}(\sigma, \mathrm{S}') &= \mathrm{lcomaj}(\sigma) - |S'| - \sum_{j \in \mathrm{Lap}(\sigma)} k \cdot |S' \cap \{1, 2, \ldots, j-1\}| \\
&= \mathrm{lcomaj}(\pi) - |T| - 1 - \sum_{j \in \mathrm{Lap}(\pi)} k \cdot |T \cap \{1, 2, \ldots, j-1\}| \\
&= \mathrm{lcomaj}(\pi) - |S| - \sum_{j \in \mathrm{Lap}(\pi)} k \cdot |S \cap \{1, 2, \ldots, j-1\}| + k(r-1) - k(s-1) \\
&= \mathrm{lcomaj}(\pi, \mathrm{S}) + k(r-s) \\
&= \mathrm{lcomaj}(\pi, \mathrm{S}) + i,
\end{aligned}$$

as desired.

Case (ii): Otherwise, we construct $\delta_{n,k,s}(i, (\pi, S))$ by the following procedure.

1. Insert k copies of n at the position labeled by i in the lcomaj-labeling of (π, S);
2. Move all the stars in each longest ascent-plateau block to the right of the last occurrence of n one (longest ascent-plateau) block to its left.

When $i \neq kr - ks$, assume that the position immediately following π_p is labeled by i in the lcomaj-labeling of (π, S). In this case, the second step is well-defined because, after the first step, $[p, p+k)$ will be an unstarred longest ascent-plateau

block. When $i = kr - ks$, the k copies of n are inserted at the start of π. In this case, the second step is well-defined since the leftmost longest ascent-plateau block (if any) is unstarred. From the construction of $\delta_{n,k,s}(i,(\pi,S))$, it is easily seen that the rightmost longest ascent-plateau block (if any) is unstarred. Hence, we have $\delta_{n,k,s}(i,(\pi,S)) \in \mathcal{X}\mathcal{Q}_n^{\mathrm{Lap}}(k,s)$.

For example, let $(\pi,S) = 16662_*4_*4_*455522_*3_*3_*311 \in \mathcal{Q}_6^{\mathrm{Lap}}(3,2)$ and $i = 5$. Clearly, the lcomaj-labeling of (π,S) is given by

$$_6 1_3 6_4 6_5 6_7 2_* 4_* 4_* 4_0 5_1 5_2 5_8 2_9 2_* 3_* 3_* 3_{10} 1_{11} 1_{12}.$$

Inserting three copies of 7 at the position labeled by 5, we obtain

$$16677762_*4_*4_*455522_*3_*3_*311.$$

Then, we get

$$\delta_{7,3,2}(5,(\pi,S)) = 166_*7_*7_*762444_*5_*5_*52233311$$

by moving all the stars in each longest ascent-plateau block to the right of the last occurrence of 7 one (longest ascent-plateau) block to its left.

Now we proceed to show that $\mathrm{lcomaj}(\delta_{n,k,s}(i,(\pi,S))) = \mathrm{lcomaj}(\pi,S) + i$. Let $(\sigma,S') = \delta_{n,k,s}(i,(\pi,S))$.

If $i = kr - ks$, then k copies of n are inserted at the start of π. The inserting of $n's$ increases $\mathrm{lcomaj}(\pi)$ by kr. Moving all the stars in s longest ascent-plateau blocks one block to its left increases the sum $\sum_{j\in \mathrm{Lap}(\pi)} k\cdot|S\cap\{1,2,\ldots,j-1\}|$ by ks. Clearly, we have $|S| = |S'|$. Hence, we have

$$\begin{aligned}
\mathrm{lcomaj}(\sigma,\mathrm{S}') &= \mathrm{lcomaj}(\sigma) - |S'| - \sum_{j\in \mathrm{Lap}(\sigma)} k\cdot|S'\cap\{1,2,\ldots,j-1\}| \\
&= \mathrm{lcomaj}(\pi) + kr - |S| - \sum_{j\in \mathrm{Lap}(\pi)} k\cdot|S\cap\{1,2,\ldots,j-1\}| - ks \\
&= \mathrm{lcomaj}(\pi,\mathrm{S}) + k(r-s) \\
&= \mathrm{lcomaj}(\pi,\mathrm{S}) + i,
\end{aligned}$$

as desired.

Now suppose that the position labeled i in the lcomaj-labeling of (π,S) is the position immediately following π_{p+j} for some $0 \le j \le k-1$ where $p \in \mathrm{Lap}(\pi)$. Suppose that there are a starred longest ascent-plateau blocks and b unstarred longest ascent-plateau blocks to the right of π_{p+j}. According to the definition of the lcomaj-labeling of (π,S), it is not difficult to check that $i = bk + j$. Inserting k copies of n at the position immediately following π_{p+j} increases $\mathrm{lcomaj}(\pi)$ by $k(a+b)+j$. Since we are moving the stars in a longest ascent-plateau blocks, the corresponding sum

$\sum_{j\in \mathrm{Lap}(\pi)} k\cdot|S\cap\{1,2,\ldots,j-1\}|$ will be increased by ka. Clearly, we have $|S|=|S'|$. Hence, we have

$$\begin{aligned}\mathrm{lcomaj}(\sigma,\mathrm{S}') &= \mathrm{lcomaj}(\sigma)-|S'|-\sum_{j\in \mathrm{Lap}(\sigma)} k\cdot|S'\cap\{1,2,\ldots,j-1\}|\\ &= \mathrm{lcomaj}(\pi)+k(a+b)+j-|S|-\sum_{j\in \mathrm{Lap}(\pi)} k\cdot|S\cap\{1,2,\ldots,j-1\}|-ka\\ &= \mathrm{lcomaj}(\pi,\mathrm{S})+kb+j\\ &= \mathrm{lcomaj}(\pi,\mathrm{S})+i,\end{aligned}$$

as desired.

Next, suppose that the position labeled i in the lcomaj-labeling of (π,S) is the position immediately following π_p for some $p\notin\overline{\mathrm{Lap}}(\pi)$. Suppose that there are a starred longest ascent-plateau blocks and b unstarred longest ascent-plateau blocks to the right of π_p, and c starred longest ascent-plateau blocks and d unstarred longest ascent-plateau blocks to the left of π_p. According to the definition of the lcomaj-labeling of (π,S), it is not difficult to check that $i=p-kc+kb$. Inserting k copies of n at the position immediately following π_p increases $\mathrm{lcomaj}(\pi)$ by $p+k(a+b)$ and increases the sum $\sum_{j\in \mathrm{Lap}(\pi)} k\cdot|S\cap\{1,2,\ldots,j-1\}|$ by kc as there are c starred longest ascent-plateau blocks to the left of the first occurrence of n. Since we are moving all the stars in a longest ascent-plateau blocks, the corresponding sum $\sum_{j\in \mathrm{Lap}(\pi)} k\cdot|S\cap\{1,2,\ldots,j-1\}|$ will be increased by ka. Clearly, we have $|S|=|S'|$. Hence, we have

$$\begin{aligned}\mathrm{lcomaj}(\sigma,\mathrm{S}') &= \mathrm{lcomaj}(\sigma)-|S'|-\sum_{j\in \mathrm{Lap}(\sigma)} k\cdot|S'\cap\{1,2,\ldots,j-1\}|\\ &= \mathrm{lcomaj}(\pi)+p+k(a+b)-|S|-\sum_{j\in \mathrm{Lap}(\pi)} k\cdot|S\cap\{1,2,\ldots,j-1\}|-kc-ka\\ &= \mathrm{lcomaj}(\pi,\mathrm{S})+p-kc+kb\\ &= \mathrm{lcomaj}(\pi,\mathrm{S})+i\end{aligned}$$

as desired.

So far, we have concluded that the map

$$\delta_{n,k,s}:\{0,1,\ldots,kn-ks-k\}\times\mathcal{Q}_{n-1}^{\mathrm{Lap}}(k,s)\longrightarrow\mathcal{X}\mathcal{Q}_n^{\mathrm{Lap}}(k,s)\bigcup\mathcal{Y}\mathcal{Q}_n^{\mathrm{Lap}}(k,s)$$

is well-defined and $\mathrm{lcomaj}(\delta_{n,k,s}(i,(\pi,S)))=\mathrm{lcomaj}(\pi,S)+i$ for any $(\pi,S)\in\mathcal{Q}_{n-1}^{\mathrm{Lap}}(k,s)$ and $0\le i\le kn-ks-k$.

In order to show that the map $\delta_{n,k,s}$ is a bijection, we describe a map

$$\bar{\delta}_{n,s,k}:\mathcal{X}\mathcal{Q}_n^{\mathrm{Lap}}(k,s)\bigcup\mathcal{Y}\mathcal{Q}_n^{\mathrm{Lap}}(k,s)\longrightarrow\{0,1,\ldots,kn-ks-k\}\times\mathcal{Q}_{n-1}^{\mathrm{Lap}}(k,s).$$

Let $(\sigma,S')\in\mathcal{X}\mathcal{Q}_n^{\mathrm{Lap}}(k,s)\bigcup\mathcal{Y}\mathcal{Q}_n^{\mathrm{Lap}}(k,s)$.

Case (i′): If $(\sigma,S')\in\mathcal{Y}\mathcal{Q}_n^{\mathrm{Lap}}(k,s)$, we construct $\bar{\delta}_{n,s,k}(\sigma,S')=(i,(\pi,S))$ by

- removing the stars in the rightmost longest ascent-plateau block,
- removing all the n's,
- moving all the stars in each longest ascent-plateau block one (longest ascent-plateau) block to its right,
- labeling the positions in the leftmost longest ascent-plateau block by stars and letting $i = kr - ks$ where $r = \mathrm{lap}(\pi)$.

Case (ii′): If $(\sigma, S') \in \mathcal{XQ}_n^{\mathrm{Lap}}(k,s)$, we construct $\bar{\delta}_{n,s,k}(\sigma, S') = (i,(\pi,S))$ by the following steps.

- Move all the stars in each longest ascent-plateau block weakly to the right of the first occurrence of n (including the longest ascent-plateau block containing n if n is a starred letter) one (longest ascent-plateau) block to its right,
- Remove the n's and record i as the label of the position where $n's$ are inserted under the lcomaj-labeling of (π, S).

Notice that the leftmost longest ascent-plateau block is starred in (π, S) and $i = kr - ks$ with $r \geq 1$ in Case (i′). Moreover, we have either $i \neq kr - ks$ or $i = kr - ks$ and the leftmost longest ascent-plateau block (if any) of (π, S) is unstarred in Case (ii′). Hence, cases (i′) and (ii′) in the construction of $\bar{\delta}_{n,k,s}$ correspond, respectively, to cases (i) and (ii) in the construction of $\delta_{n,k,s}$. This implies that the maps $\delta_{n,s,k}$ and $\bar{\delta}_{n,s,k}$ are inverses of each other, and therefore the map $\delta_{n,s,k}$ is indeed a bijection, completing the proof. ∎

Lemma 2.6 *For $n \geq 2$ and $s \geq 1$, there is a bijection*

$$\delta^*_{n,k,s} : \{0,1,\ldots,kn-ks-1\} \times \mathcal{Q}^{\mathrm{Lap}}_{n-1}(k,s-1) \longrightarrow \mathcal{ZQ}^{\mathrm{Lap}}_n(k,s)$$

such that $\mathrm{lcomaj}(\delta^*_{n,k,s}(i,(\pi,S))) = \mathrm{lcomaj}(\pi,S) + i$ *for any* $(\pi,S) \in \mathcal{Q}^{\mathrm{Lap}}_{n-1}(k,s-1)$ *and* $0 \leq i \leq kn - ks - 1$.

Proof. First, we give a description of the map $\delta^*_{n,k,s}$. Let $(i,(\pi,S)) \in \{0,1,\ldots,kn - ks - 1\} \times \mathcal{Q}^{\mathrm{Lap}}_{n-1}(k,s-1)$ with $\mathrm{lap}(\pi) = r$. In the following, we demonstrate how to generate $\delta^*_{n,k,s}(i,(\pi,S))$ and prove that $\mathrm{lcomaj}(\delta^*_{n,k,s}(i,(\pi,S))) = \mathrm{lcomaj}(\pi,S) + i$.
Case (i): $i = kr - ks + k - 1$ with $r \geq 1$ and the leftmost longest ascent-plateau block is starred. We construct $(\sigma, S') = \delta^*_{n,k,s}(i,(\pi,S))$ by the following procedure.

1. Remove the stars in the leftmost longest ascent-plateau block;

2. Move all the stars in each longest ascent-plateau block one (longest ascent-plateau) block to its left;

3. Label the positions in the rightmost longest ascent-plateau block by stars;

4. Let (π, T) be the resulting starred permutation;

5. Let (σ, S') be the starred permutation obtained from $\delta_{n,k,s-1}(0,(\pi,T))$ by labeling the positions in its rightmost longest ascent-plateau block by stars.

The second step is well-defined since the leftmost longest ascent-plateau block will be unstarred after the first step. Moreover, the third step is well-defined because the rightmost longest ascent-plateau block will be unstarred after the second step. Recall that $\mathrm{lap}(\pi) = r$ and $|T| = |S| = s-1$. Since $i = kr - ks + k - 1 \geq 0$, we have $kr - k(s-1) \neq 0$. From the construction of $\delta_{n,k,s-1}$, we have $\delta_{n,k,s-1}(0,(\pi,T)) \in \mathcal{XQ}_n^{\mathrm{Lap}}(k, s-1)$. that is, $\delta_{n,k,s-1}(0,(\pi,T))$ has $s-1$ starred longest ascent-plateau blocks and the rightmost longest ascent-plateau block of $\delta_{n,k,s-1}(0,(\pi,T))$ is unstarred. This guarantees that the fifth step is feasible and the resulting starred permutation (σ, S') has s starred longest ascent-plateau blocks. As the rightmost longest ascent-plateau block is starred, we have $(\sigma, S') \notin \mathcal{XQ}_n^{\mathrm{Lap}}(k, s)$. Assume that the position immediately following π_p is labeled by 0 under the lcomaj-labeling of (π, T). As the rightmost longest ascent-plateau block is starred in (π, T), $[p, p+k)$ must not be the rightmost longest ascent-plateau block of (σ, S'). This ensures that (σ, S') does not verify property (i) of $\mathcal{YQ}_n^{\mathrm{Lap}}(k, s)$, which implies that $(\sigma, S') \notin \mathcal{YQ}_n^{\mathrm{Lap}}(k, s)$. Therefore, we have $(\sigma, S') \in \mathcal{ZQ}_n^{\mathrm{Lap}}(k, s)$ as desired, that is, the map $\delta^*_{n,k,s}$ is well-defined in this case.

For example, let $(\pi, S) = 12_*4_*4_*455522_*3_*3_*311 \in \mathcal{Q}_5^{\mathrm{Lap}}(3, 2)$ and let $i = 2$. Clearly, the lcomaj-labeling of (π, S) is given by

$$ {}_3 1_4 2_* 4_* 4_* 4_0 5_1 5_2 5_5 2_6 2_* 3_* 3_* 3_7 1_8 1_9. $$

Removing the stars at the positions in the leftmost longest ascent-plateau block, we obtain $1244455522_*3_*3_*311$. Then, we get

$$ (\pi, T) = 12444_*5_*5_*522_*3_*3_*311 $$

by moving all the stars in each longest ascent-plateau block one (longest ascent-plateau) block to its left and labeling the rightmost longest ascent-plateau block by stars. Finally, we obtain

$$ \delta^*_{6,3,3}(2,(\pi,S)) = 12_*6_*6_*6444_*5_*5_*522_*3_*3_*311 $$

by labeling the position in the rightmost longest ascent-plateau block of

$$\delta_{6,3,2}(0,(\pi,T)) = 12_*6_*6_*6444_*5_*5_*52233311$$

by stars.

Now we proceed to show that $\mathrm{lcomaj}(\sigma,S') = \mathrm{lcomaj}(\pi,S)+i$. Recall that

$$\mathrm{lcomaj}(\pi,\mathrm{S}) = \mathrm{lcomaj}(\pi) - |S| - \sum_{j\in \mathrm{Lap}(\pi)} k\cdot|S\cap\{1,2,\ldots,j-1\}|.$$

Removing the stars in the leftmost longest ascent-plateau block decreases $|S|$ by 1 and decreases the sum $\sum_{j\in \mathrm{Lap}(\pi)} k\cdot|S\cap\{1,2,\ldots,j-1\}|$ by $k(r-1)$. After moving all the stars in $s-2$ longest ascent-plateau blocks one (longest ascent-plateau) block to its left the sum $\sum_{j\in \mathrm{Lap}(\pi)} k\cdot|S\cap\{1,2,\ldots,j-1\}|$ will be increased by $k(s-2)$. Labeling the positions in the rightmost longest ascent-plateau block by stars increases $|S|$ by 1 and does not affect the sum $\sum_{j\in \mathrm{Lap}(\pi)} k\cdot|S\cap\{1,2,\ldots,j-1\}|$. Hence, we have

$$|T|+\sum_{j\in \mathrm{Lap}(\pi)} k\cdot|T\cap\{1,2,\ldots,j-1\}| = k(s-2)-k(r-1)+|S|+\sum_{j\in \mathrm{Lap}(\pi)} k\cdot|S\cap\{1,2,\ldots,j-1\}|.$$

Then, we have

$$\begin{aligned}
\mathrm{lcomaj}(\pi,\mathrm{T}) &= \mathrm{lcomaj}(\pi) - |T| - \sum_{j\in \mathrm{Lap}(\pi)} k\cdot|T\cap\{1,2,\ldots,j-1\}| \\
&= \mathrm{lcomaj}(\pi) - k(s-2) + k(r-1) - |S| - \sum_{j\in \mathrm{Lap}(\pi)} k\cdot|S\cap\{1,2,\ldots,j-1\}| \\
&= \mathrm{lcomaj}(\pi,\mathrm{S}) + kr - ks + k
\end{aligned}$$

Let $(\tau,T') = \delta_{n,k,s-1}(0,(\pi,T))$. By Lemma 2.5, we have $\mathrm{lcomaj}(\tau,T') = \mathrm{lcomaj}(\pi,T)$. Then (σ,S') is the starred permutation that arises by labeling the positions in the rightmost longest ascent-plateau block of (τ,T'). Clearly, we have $\sigma=\tau$, $|S'| = |T'|+1$ and

$$\sum_{j\in \mathrm{Lap}(\tau)} k\cdot|T'\cap\{1,2,\ldots,j-1\}| = \sum_{j\in \mathrm{Lap}(\sigma)} k\cdot|S'\cap\{1,2,\ldots,j-1\}|.$$

Hence, we have

$$\begin{aligned}
\mathrm{lcomaj}(\sigma,S') &= \mathrm{lcomaj}(\tau,T') - 1 \\
&= \mathrm{lcomaj}(\pi,T) - 1 \\
&= \mathrm{lcomaj}(\pi,\mathrm{S}) + kr - ks + k - 1 \\
&= \mathrm{lcomaj}(\pi,\mathrm{S}) + i,
\end{aligned}$$

as desired.

Case (ii): Otherwise, let $\delta^*_{n,k,s}(i,(\pi,S))$ be a starred permutation obtained from $\delta_{n,k,s-1}(i+1,(\pi,S))$ by labeling the positions in the rightmost longest ascent-plateau block by stars.

For example, let $(\pi,S) = 16662_*4_*4_*455522_*3_*3_*311 \in \mathcal{Q}_6^{\mathrm{Lap}}(3,2)$ and $i = 4$. Then, we get

$$\delta^*_{7,3,3}(4,(\pi,S)) = 166_*7_*7_*762444_*5_*5_*522_*3_*3_*311$$

by labeling the positions in the rightmost longest ascent-plateau block of

$$\delta_{7,3,2}(5,(\pi,S)) = 166_*7_*7_*762444_*5_*5_*52233311$$

by stars.

As we have either $i+1 \neq kr-k(s-1)$ or $i+1 = kr-k(s-1)$ and the leftmost longest ascent-plateau block (if any) is unstarred, we have $\delta_{n,k,s-1}(i+1,(\pi,S)) \in \mathcal{X}\mathcal{Q}_n^{\mathrm{Lap}}(k,s-1)$. Recall that in the construction of $\delta_{n,k,s-1}(i+1,(\pi,S))$, the k copies of n are inserted at the position labeled $i+1$ under the lcomaj-labeling of (π,S). If $i+1 \neq kr-k(s-1)$, then n's are not inserted at the start of π. The insertion of n's ensures that there is at least one longest ascent-plateau block in $\delta_{n,k,s-1}(i+1,(\pi,S))$. If $i+1 = kr-k(s-1)$, then n's are inserted at the start of π. As $i \geq 0$, we have $\mathrm{lap}(\pi) = r > 0$. This implies that there is at least one longest ascent-plateau block in $\delta_{n,k,s-1}(i+1,(\pi,S))$. Notice that the rightmost longest ascent-plateau block of any starred permutations in $\delta_{n,k,s-1}(i+1,(\pi,S))$ is unstarred. Hence, the map $\delta^*_{n,k,s}$ is feasible.

Let $(\tau,T) = \delta_{n,k,s-1}(i+1,(\pi,S))$ and let $(\sigma,S') = \delta^*_{n,k,s}(i+1,(\pi,S))$. We proceed to prove that $(\sigma,S') \in \mathcal{Z}\mathcal{Q}_n^{\mathrm{Lap}}(k,s)$. Clearly, we have $\sigma = \tau$. Observe that the rightmost longest ascent-plateau block is starred in (σ,S'). This implies that $(\sigma,S') \notin \mathcal{X}\mathcal{Q}_n^{\mathrm{Lap}}(k,s)$.

Recall that in the construction of (τ,T), τ is obtained from π by inserting k copies of n at the position labeled $i+1$ under the lcomaj-labeling of (π,S). Since $i+1 > 0$, it is easy to check that τ does not verify property (i) or (iii) of $\mathcal{Y}\mathcal{Q}_n^{\mathrm{Lap}}(k,s)$. As $\sigma = \tau$, we have that $(\sigma,S') \notin \mathcal{Y}\mathcal{Q}_n^{\mathrm{Lap}}(k,s)$. In conclusion, we have $(\sigma,S') \in \mathcal{Z}\mathcal{Q}_n^{\mathrm{Lap}}(k,s)$, that is, the map $\delta^*_{n,k,s}$ is well-defined in this case.

By Lemma 2.5, we have $\mathrm{lcomaj}(\delta_{n,k,s-1}(i+1,(\pi,S))) = \mathrm{lcomaj}(\pi,S)+i+1$. Recall that $\delta^*_{n,k,s}(i,(\pi,S))$ is obtained from $\delta_{n,k,s-1}(i+1,(\pi,S))$ by labeling the positions in the rightmost longest ascent-plateau block by stars. Clearly, this process decreases $\mathrm{lcomaj}(\delta_{n,k,s-1}(i+1,(\pi,S)))$ by one. Hence, we have $\mathrm{lcomaj}(\delta^*_{n,k,s}(i,(\pi,S))) = \mathrm{lcomaj}(\pi,S)+i$.

In order to show the map $\delta^*_{n,k,s}$ is a bijection, we give a description of a map $\bar{\delta}^*_{n,k,s} : \mathcal{Z}\mathcal{Q}_n^{\mathrm{Lap}}(k,s) \longrightarrow \{0,1,\ldots,kn-ks-1\} \times \mathcal{Q}_{n-1}^{\mathrm{Lap}}(k,s-1)$.

Let $(\sigma, S') \in \mathcal{ZQ}_n^{\mathrm{Lap}}(k,s)$. In the following, we proceed to demonstrate how to generate $(i,(\pi,S)) = \bar{\delta}^*_{n,k,s}(\sigma,S')$.

Let $(\tau,T) \in \mathcal{Q}_{n-1}^{\mathrm{Lap}}(k,s-1)$ be the unique starred permutation generated from (σ,S') by the following procedure.

- Remove all the stars in the rightmost longest ascent-plateau block.
- Move all the stars in each longest ascent-plateau block weakly to the right of the first occurrence of n (including the longest ascent-plateau block containing n if n is a starred letter) one (longest ascent-plateau) block to its right.
- Remove all the $n's$.

Record i' as the label of the position where $n's$ are inserted under the lcomaj-labeling of (τ,T).

Case (i′): If $i'=0$, then we generate $(i,(\pi,S))$ from (τ,T) by

- removing all the stars in the rightmost longest ascent-plateau block,
- moving all the stars in each longest ascent-plateau block one (longest ascent-plateau) block to its right,
- labeling the positions in the leftmost longest ascent-plateau block by stars and letting $i = kr-ks+k-1$ where $r=\mathrm{lap}(\pi)$.

Case (ii′): Otherwise, set $(\pi,S)=(\tau,T)$ and $i=i'-1$.

It is obvious that the leftmost longest ascent-plateau block is starred in (π,S) and $i=kr-ks+k-1$ in Case (i′). Moreover, we have either $i \neq kr-ks+k-1$ or $i=kr-ks+k-1$ and the leftmost longest ascent-plateau block of (π,S) is unstarred in Case (ii′). Hence, cases (i′) and (ii′) in the construction of $\bar{\delta}^*_{n,k,s}$ correspond, respectively, to cases (i) and (ii) in the construction of $\delta^*_{n,k,s}$. This implies that the maps $\delta^*_{n,s,k}$ and $\bar{\delta}^*_{n,s,k}$ are inverses of each other, and therefore the map $\delta^*_{n,s,k}$ is indeed a bijection, completing the proof. ∎

Lemma 2.7 *For $n \geq 2$, the polynomial $\tilde{Q}^{\mathrm{Lap}}_{n,k,s}(q)$ equals 0 when $s<0$ or $s>n-1$ and*

$$\tilde{Q}^{\mathrm{Lap}}_{n,k,s}(q) = [kn-ks-k+1]_q \tilde{Q}^{\mathrm{Lap}}_{n-1,k,s}(q) + [kn-ks]_q \tilde{Q}^{\mathrm{Lap}}_{n-1,k,s-1}(q)$$

otherwise.

Proof. It follows from Lemmas 2.5 and 2.6 that

$$\begin{aligned}
\tilde{Q}^{\mathrm{Lap}}_{n,k,s}(q) &= \sum_{(\pi,S)\in\mathcal{Q}^{\mathrm{Lap}}_n(k,s)} q^{\mathrm{lcomaj}(\pi,\mathrm{S})} \\
&= \sum_{(\pi,S)\in\mathcal{X}\mathcal{Q}^{\mathrm{Lap}}_n(k,s)\cup\mathcal{Y}\mathcal{Q}^{\mathrm{Lap}}_n(k,s)} q^{\mathrm{lcomaj}(\pi,\mathrm{S})} + \sum_{(\pi,S)\in\mathcal{Z}\mathcal{Q}^{\mathrm{Lap}}_n(k,s)} q^{\mathrm{lcomaj}(\pi,\mathrm{S})} \\
&= \sum_{i=0}^{kn-ks-k} \sum_{(\pi,S)\in\mathcal{Q}^{\mathrm{Lap}}_{n-1}(k,s)} q^{i+\mathrm{lcomaj}(\pi,\mathrm{S})} + \sum_{i=0}^{kn-ks-1} \sum_{(\pi,S)\in\mathcal{Q}^{\mathrm{Lap}}_{n-1}(k,s-1)} q^{i+\mathrm{lcomaj}(\pi,\mathrm{S})} \\
&= [kn-ks-k+1]_q \tilde{Q}^{\mathrm{Lap}}_{n-1,k,s}(q) + [kn-ks]_q \tilde{Q}^{\mathrm{Lap}}_{n-1,k,s-1}(q),
\end{aligned}$$

as desired. ∎

Proof of Conjecture 1.1. By Lemmas 2.3 and 2.7, we have $\tilde{Q}^{\mathrm{Lap}}_{n,k,s}(q) = Q^{\mathrm{Lap}}_{n,k,s}(q)$ since $\tilde{Q}^{\mathrm{Lap}}_{n,k,s}(q)$ satisfies the same recurrence and initial conditions as $Q^{\mathrm{Lap}}_{n,k,s}(q)$. Then Conjecture 1.1 follows from Lemmas 2.1 and 2.4. This completes the proof. ∎

3 Proof of Theorem 1.2

This section is devoted to the proof of Theorem 1.2. Since the main idea of the proof is very similar to the one for Conjecture 1.1, some detailed proofs will be omitted.

Given a permutation $\pi \in \mathcal{Q}_n(k)$ and a set $S \subseteq \mathrm{Pp}(\pi)$, the pair (π, S) is said to be a *plateau-peak starred k-Stirling permutation* if for each $i \in S$, we put a star at each position after π_j for all $j \in \{i, i+1, \ldots, i+k\}$. For example, for $\pi = 125552444233311$ and $S = \{2, 10\}$, we have $(\pi, S) = 12_*5_*5_*5_*24442_*3_*3_*3_*11$.

Define

$$\mathcal{Q}^{\mathrm{Pp}}_n(k) = \{(\pi, S) \mid \pi \in \mathcal{Q}_n(k), S \subseteq \mathrm{Pp}(\pi)\}$$

and

$$\mathcal{Q}^{\mathrm{Pp}}_n(k,s) = \{(\pi, S) \mid (\pi, S) \in \mathcal{Q}^{\mathrm{Pp}}_n(k), |S| = s\}.$$

For any permutation $(\pi, S) \in \mathcal{Q}^{\mathrm{Pp}}_n(k)$, a letter π_i is said to be a *starred* (resp. *unstarred*) letter if the position before π_i is starred (resp. unstarred). Define the *last inversion* of (π, S), denoted by $\mathrm{linv}(\pi, \mathrm{S})$, to be

$$\mathrm{linv}(\pi, \mathrm{S}) = \mathrm{linv}(\pi) - \sum_{j\in S} (k+1) \cdot \mathrm{linv}^{\square, \mathrm{j}+1}(\pi).$$

For example, if let $(\pi, S) = 12_*5_*5_*5_*24442_*3_*3_*3_*11 \in \mathcal{Q}^{\mathrm{Pp}}_5(3)$, then

$$\mathrm{Pp}(\pi) = \{2, 6, 10\},\ S = \{2, 10\}$$

and

$$\begin{aligned}\mathrm{linv}(\pi,\mathrm{S}) &= \mathrm{linv}(\pi)-\sum_{j\in S}(k+1)\cdot\mathrm{linv}^{\square,\mathrm{j}+1}(\pi)\\ &= 20-4\cdot\mathrm{linv}^{\square,3}(\pi)-4\cdot\mathrm{linv}^{\square,11}(\pi)\\ &= 20-0-4\times 2\\ &= 12.\end{aligned}$$

Define

$$Q^{\mathrm{Pp}}_{n,k,s}(q)=\sum_{(\pi,S)\in\mathcal{Q}^{\mathrm{Pp}}_n(k,s)} q^{\mathrm{linv}(\pi,\mathrm{S})}.$$

By a similar argument as Lemma 2.1, we have the following lemma for the left-hand side of Theorem 1.2.

Lemma 3.1 *We have*

$$\sum_{\pi\in\mathcal{Q}_n(k)} q^{\mathrm{linv}(\pi)}\prod_{j\in\mathrm{Pp}(\pi)}\left(1+\frac{z}{q^{(k+1)\cdot\mathrm{linv}^{\square,j+1}(\pi)}}\right)=\sum_{s\geq 0}Q^{\mathrm{Pp}}_{n,k,s}(q)z^s. \tag{3.1}$$

We shall derive a recurrence relation for $Q^{\mathrm{Pp}}_{n,k,s}(q)$. To this end, we give a linv-labeling for permutations in $\mathcal{Q}^{\mathrm{Pp}}_n(k)$. Let $(\pi,S)\in\mathcal{Q}^{\mathrm{Pp}}_n(k,s)$, the linv-labeling of (π,S) is obtained by labeling the unstarred positions of π by $0,1,\ldots,kn-(k+1)s$ from right to left. For example, for $(\pi,S)=12_*5_*5_*5_*24442_*3_*3_*3_*11$, the linv-labeling of (π,S) is

$$_7\mathbf{1}_6\mathbf{2}_*\mathbf{5}_*\mathbf{5}_*\mathbf{5}_*\mathbf{2}_5\mathbf{4}_4\mathbf{4}_3\mathbf{4}_2\mathbf{2}_*\mathbf{3}_*\mathbf{3}_*\mathbf{3}_*\mathbf{1}_1\mathbf{1}_0,$$

where the labels of the positions are written as subscripts.

Denote by $\bar{\mathcal{Q}}^{\mathrm{Pp}}_n(k,s)$ (resp. $\hat{\mathcal{Q}}^{\mathrm{Pp}}_n(k,s)$) the set of starred permutations $(\pi,S)\in\mathcal{Q}^{\mathrm{Pp}}_n(k,s)$ such that all the n's are unstarred (resp. starred) in (π,S). By similar analysis in the construction of the maps ϕ and ϕ^*, we can define the following two maps for the generation of the starred permutations in $\mathcal{Q}^{\mathrm{Pp}}_n(k,s)$. Define

$$\psi_{n,k,s}:\{0,1,\ldots,kn-(k+1)s-k\}\times\mathcal{Q}^{\mathrm{Pp}}_{n-1}(k,s)\longrightarrow\bar{\mathcal{Q}}^{\mathrm{Pp}}_n(k,s)$$

by mapping the pair $(i,(\pi,S))$ to the starred permutation by inserting k copies of n at the position in (π,S) which is labeled by i in the linv-labeling of (π,S). For example, we have

$$\psi_{6,3,2}(5,(\pi,S))=12_*5_*5_*5_*26664442_*3_*3_*3_*11,$$

where $(\pi,S)=12_*5_*5_*5_*24442_*3_*3_*3_*11$.

Define

$$\psi^*_{n,k,s} : \{0, 1, \ldots, kn - (k+1)s\} \times \mathcal{Q}^{\mathrm{Pp}}_{n-1}(k, s-1) \longrightarrow \hat{\mathcal{Q}}^{\mathrm{Pp}}_n(k, s)$$

by mapping the pair $(i, (\pi, S))$ to the starred permutation by inserting k copies of n at the position in (π, S) which is labeled by i in the linv-labeling of (π, S), labeling the position before each n by a star, and labeling the position after the last occurrence of n by a star. Here we should note that we cannot insert k copies of n at the start of π by the definition of the starred permutations. For example, we have

$$\psi^*_{6,3,3}(5, (\pi, S)) = 12_*5_*5_*5_*2_*6_*6_*6_*4442_*3_*3_*3_*11,$$

where $(\pi, S) = 12_*5_*5_*5_*24442_*3_*3_*3_*11$. One can easily check that both $\psi_{n,k,s}$ and $\psi^*_{n,k,s}$ are bijections.

The following lemma can be verified by the same reasoning as in the proof of Lemma 2.2.

Lemma 3.2 *We have*

- *for* $(\pi, S) \in \mathcal{Q}^{\mathrm{Pp}}_{n-1}(k, s)$ *and* $0 \le i \le kn - (k+1)s - k$,

$$\mathrm{linv}(\psi_{n,k,s}(i, (\pi, S))) = \mathrm{linv}(\pi, S) + i;$$

- *for* $(\pi, S) \in \mathcal{Q}^{\mathrm{Pp}}_{n-1}(k, s-1)$ *and* $0 \le i \le kn - (k+1)s$,

$$\mathrm{linv}(\psi^*_{n,k,s}(i, (\pi, S))) = \mathrm{linv}(\pi, S) + i.$$

Relying on Lemma 3.2, one can verify the following result by the same reasoning as in the proof of Lemma 2.3.

Lemma 3.3 *For* $n \ge 2$, *the polynomial* $Q^{\mathrm{Pp}}_{n,k,s}(q)$ *equals* 0 *when* $s < 0$ *or* $s > n - 1$ *and*

$$Q^{\mathrm{Pp}}_{n,k,s}(q) = [kn - (k+1)s - k + 1]_q Q^{\mathrm{Pp}}_{n-1,k,s}(q) + [kn - (k+1)s + 1]_q Q^{\mathrm{Pp}}_{n-1,k,s-1}(q)$$

otherwise.

Next, we consider the right-hand side of Theorem 1.2. For $\pi \in \mathcal{Q}_n(k)$, if $i \in \mathrm{Pp}(\pi)$, the set $[i, i+k] = \{i, i+1, \ldots, i+k\}$ is called a *plateau-peak block* of π. Let $\overline{\mathrm{Pp}}(\pi) = \bigcup_{i \in \mathrm{Pp}(\pi)} [i, i+k]$. For $(\pi, S) \in \mathcal{Q}^{\mathrm{Pp}}_n(k)$, if $i \in S$, we say that the plateau-peak

block $[i, i+k]$ is starred. Note that all the positions in a starred plateau-peak block are starred.

By similar analysis for the right-hand side of Conjecture 1.1, in order to give a combinatorial interpretation of the right-hand side of Theorem 1.2, we define the pp-maj of $(\pi, S) \in \mathcal{Q}_n^{\mathrm{Pp}}(k)$, denoted by pp-maj$(\pi, S)$, to be

$$\text{pp-maj}(\pi, S) = \text{pp-maj}(\pi) - |S| - \sum_{j \in \mathrm{Pp}(\pi)} (k+1) \cdot |S \cap \{1, 2, \ldots, j-1\}|.$$

For example, if let $(\pi, S) = 12_*5_*5_*5_*24442_*3_*3_*3_*11 \in \mathcal{Q}_5^{\mathrm{Pp}}(3)$, then

$$\mathrm{Pp}(\pi) = \{2, 6, 10\},\ S = \{2, 10\}$$

and

$$\begin{aligned} \text{pp-maj}(\pi, S) &= \text{pp-maj}(\pi) - |S| - \sum_{j \in \mathrm{Pp}(\pi)} (k+1) \cdot |S \cap \{1, 2, \ldots, j-1\}| \\ &= 21 - 2 - 0 - 4 - 4 \\ &= 11. \end{aligned}$$

Define

$$\tilde{Q}_{n,k,s}^{\mathrm{Pp}}(q) = \sum_{(\pi,S) \in \mathcal{Q}_n^{\mathrm{Pp}}(k,s)} q^{\text{pp-maj}(\pi,S)}.$$

The following lemma can be verified by similar arguments as in the proof of Lemma 2.4.

Lemma 3.4 *We have*

$$\sum_{\pi \in \mathcal{Q}_n(k)} q^{\text{pp-maj}(\pi)} \prod_{j=1}^{\mathrm{pp}(\pi)} \left(1 + \frac{z}{q^{(k+1)(j-1)+1}}\right) = \sum_{s \geq 0} \tilde{Q}_{n,k,s}^{\mathrm{Pp}}(q) z^s.$$

In the rest of this section, we will devote to deriving a recurrence relation for $\tilde{Q}_{n,k,s}^{\mathrm{Pp}}(q)$, which turns out to be the same as the one for $Q_{n,k,s}^{\mathrm{Pp}}(q)$. This allows us to prove Theorem 1.2 immediately.

We give a pp-maj-labeling for permutations in $\mathcal{Q}_n^{\mathrm{Pp}}(k)$. For $(\pi, S) \in \mathcal{Q}_n^{\mathrm{Pp}}(k)$ with $|S| = s$ and $\mathrm{pp}(\pi) = r$, the pp-maj-labeling of (π, S) is obtained as follows.

- Label the positions in the rightmost unstarred plateau-peak block from left to right by $0, 1, \ldots, k$, label the positions in the second to the rightmost unstarred plateau-peak block from left to right by $k+1, k+2, \ldots, 2k+1$, and so on.
- Label the remaining unstarred positions from left to right by $(k+1)(r-s), (k+1)(r-s)+1, \ldots, kn-(k+1)s$.

Since the leftmost position is always unstarred, the label of this position is $(k+1)(r-s)$.

For example, given $(\pi, S) = 12_*5_*5_*5_*24442_*3_*3_*3_*11$, the pp-maj-labeling of (π, S) is

$$_4 1_5 2_* 5_* 5_* 5_* 2_0 4_1 4_2 4_3 2_* 3_* 3_* 3_* 1_6 1_7,$$

where the labels of the positions are written as subscripts.

Let $\mathcal{XQ}_n^{\mathrm{Pp}}(k,s)$ denote the set of starred permutations $(\pi, S) \in \mathcal{Q}_n^{\mathrm{Pp}}(k,s)$ whose rightmost plateau-peak block (if any) is unstarred. Let $\mathcal{YQ}_n^{\mathrm{Pp}}(k,s)$ denote the set of starred permutations $(\sigma, S) \in \mathcal{Q}_n^{\mathrm{Pp}}(k,s)$ satisfying that

(i) $[p, p+k]$ is the rightmost plateau-peak block of σ where $\sigma_{p+i} = n$ for all $1 \le i \le k$;

(ii) the rightmost plateau-peak block $[p, p+k]$ is starred;

(iii) $\mathrm{Pp}(\pi) = \mathrm{Pp}(\sigma)$ where π is obtained from σ by removing all the $n's$.

Define

$$\mathcal{ZQ}_n^{\mathrm{Pp}}(k,s) = \mathcal{Q}_n^{\mathrm{Pp}}(k,s) - \mathcal{XQ}_n^{\mathrm{Pp}}(k,s) - \mathcal{YQ}_n^{\mathrm{Pp}}(k,s).$$

Lemma 3.5 *For $n \ge 2$ and $s \ge 0$, there is a bijection*

$$\gamma_{n,k,s} : \{0, 1, \ldots, kn-(k+1)s-k\} \times \mathcal{Q}_{n-1}^{\mathrm{Pp}}(k,s) \longrightarrow \mathcal{XQ}_n^{\mathrm{Pp}}(k,s) \bigcup \mathcal{YQ}_n^{\mathrm{Pp}}(k,s)$$

such that $\text{pp-maj}(\gamma_{n,k,s}(i,(\pi,S))) = \text{pp-maj}(\pi,S) + i$ *for any* $(\pi,S) \in \mathcal{Q}_{n-1}^{\mathrm{Pp}}(k,s)$ *and* $0 \le i \le kn-(k+1)s-k$.

Proof. First, we give a description of the map $\gamma_{n,k,s}$. Let $(i,(\pi,S)) \in \{0,1,\ldots,kn-(k+1)s-k\} \times \mathcal{Q}_{n-1}^{\mathrm{Pp}}(k,s)$ with $\mathrm{pp}(\pi) = r$. Now we shall demonstrate how to construct $\gamma_{n,k,s}(i,(\pi,S))$ and show that $\text{pp-maj}(\gamma_{n,k,s}(i,(\pi,S))) = \text{pp-maj}(\pi,S) + i$ by distinguishing the following two cases.

Case (i): $i = (k+1)(r-s)$ with $r \ge 1$ and the leftmost plateau-peak block of (π, S) is starred. We construct $\gamma_{n,k,s}(i,(\pi,S))$ by the following procedure.

1. Remove the stars in the leftmost plateau-peak block;

2. Move all the stars in each plateau-peak block one (plateau-peak) block to its left;

3. Let (π, T) be the resulting starred permutation and let $[p, p+k]$ be the rightmost plateau-peak block in (π, T);

4. Insert k copies of n at the position immediately following π_p, label the position before each n by a star, and label the position after the last occurrence of n by a star.

By similar analysis in Case (i) of the map $\delta_{n,k,s}$, it is not difficult to verify that $\gamma_{n,k,s}$ is well-defined and $\gamma_{n,k,s}(i,(\pi,S)) \in \mathcal{YQ}_n^{\mathrm{Pp}}(k,s)$.

For example, if let $(\pi,S) = 12_*5_*5_*5_*24442_*3_*3_*3_*11$, then

$$\gamma_{6,3,2}(4,(\pi,S)) = 125552_*4_*4_*4_*2_*6_*6_*6_*33311.$$

Recall that

$$\text{pp-maj}(\pi,S) = \text{pp-maj}(\pi) - |S| - \sum_{j\in \mathrm{Pp}(\pi)} (k+1)\cdot |S\cap\{1,2,\ldots,j-1\}|.$$

Removing the stars in the leftmost plateau-peak block decreases $|S|$ by 1 and decreases the sum $\sum_{j\in \mathrm{Pp}(\pi)} (k+1)\cdot |S\cap\{1,2,\ldots,j-1\}|$ by $(k+1)(r-1)$. Moving all the stars in the $s-1$ plateau-peak blocks one (plateau-peak) block to its left, the sum $\sum_{j\in \mathrm{Pp}(\pi)} (k+1)\cdot |S\cap\{1,2,\ldots,j-1\}|$ will be increased by $(k+1)(s-1)$. Hence, we have

$$\begin{aligned}
&|T| + \sum_{j\in \mathrm{Pp}(\pi)} (k+1)\cdot |T\cap\{1,2,\ldots,j-1\}| \\
&= (k+1)(s-1) - (k+1)(r-1) + |S| - 1 + \sum_{j\in \mathrm{Pp}(\pi)} (k+1)\cdot |S\cap\{1,2,\ldots,j-1\}|.
\end{aligned}$$

Let $(\sigma,S') = \gamma_{n,k,s}(i,(\pi,S))$. Observe that (σ,S') arises from (π,T) by inserting k copies of n at the position immediately following π_p, labeling the position before each n by a star, and labeling the position after the last occurrence of n by a star. Clearly, this process does not affect the sum $\sum_{j\in \mathrm{Pp}(\pi)} (k+1)\cdot |T\cap\{1,2,\ldots,j-1\}|$ as $[p,p+k]$ is the rightmost plateau-peak block of π. Moreover, we have $S' = T\cup\{p\}$, $\mathrm{Pp}(\pi) = \mathrm{Pp}(\sigma)$, $\mathrm{pp}(\pi_{(n)}) = 0$, and $\mathrm{pp}(\pi_{(i)}) = \mathrm{pp}(\sigma_{(i)})$ for all $1\le i\le n-1$. Hence, we have

$$\begin{aligned}
\text{pp-maj}(\sigma,S') &= \text{pp-maj}(\sigma) - |S'| - \sum_{j\in \mathrm{Pp}(\sigma)} (k+1)\cdot |S'\cap\{1,2,\ldots,j-1\}| \\
&= \text{pp-maj}(\pi) - |T| - 1 - \sum_{j\in \mathrm{Pp}(\pi)} (k+1)\cdot |T\cap\{1,2,\ldots,j-1\}| \\
&= \text{pp-maj}(\pi) - |S| - \sum_{j\in \mathrm{Pp}(\pi)} (k+1)\cdot |S\cap\{1,2,\ldots,j-1\}| + (k+1)(r-s) \\
&= \text{pp-maj}(\pi,S) + (k+1)(r-s) \\
&= \text{pp-maj}(\pi,S) + i,
\end{aligned}$$

as desired.

Case (ii): Otherwise, we construct $\gamma_{n,k,s}(i,(\pi,S))$ by the following procedure.

1. Insert k copies of n at the position labeled by i in the pp-maj-labeling of (π, S);

2. Move all the stars in each plateau-peak block to the right of the last occurrence of n one (plateau-peak) block to its left.

By the same reasoning as in the proof of Case (ii) of the map $\delta_{n,k,s}$, one can verify that the map $\gamma_{n,k,s}$ is well-defined and we have $\gamma_{n,k,s}(i,(\pi,S)) \in \mathcal{XQ}_n^{\mathrm{Pp}}(k,s)$.

For example, if let $(\pi, S) = 12_*5_*5_*5_*24442_*3_*3_*3_*11$, then

$$\gamma_{6,3,2}(2,(\pi,S)) = 12_*5_*5_*5_*244_*6_*6_*6_*4233311.$$

Now we proceed to show that $\text{pp-maj}(\gamma_{n,k,s}(i,(\pi,S))) = \text{pp-maj}(\pi,S) + i$. Let $(\sigma, S') = \gamma_{n,k,s}(i,(\pi,S))$.

If $i = (k+1)(r-s)$, then k copies of n are inserted at the start of π. The inserting of $n's$ increases $\text{pp-maj}(\pi)$ by $(k+1)r$. Moving all the stars in s plateau-peak blocks one block to its left increases the sum $\sum_{j\in \mathrm{Pp}(\pi)} (k+1)\cdot|S\cap\{1,2,\ldots,j-1\}|$ by $(k+1)s$. Hence, we have

$$\begin{aligned}
\text{pp-maj}(\sigma,S') &= \text{pp-maj}(\sigma) - |S'| - \sum_{j\in \mathrm{Pp}(\sigma)} (k+1)\cdot|S'\cap\{1,2,\ldots,j-1\}| \\
&= \text{pp-maj}(\pi) + (k+1)r - |S| - \sum_{j\in \mathrm{Pp}(\pi)} (k+1)\cdot|S\cap\{1,2,\ldots,j-1\}| - (k+1)s \\
&= \text{pp-maj}(\pi,S) + (k+1)(r-s) \\
&= \text{pp-maj}(\pi,S) + i,
\end{aligned}$$

as desired.

Now suppose that the position labeled i in the lcomaj-labeling of (π, S) is the position immediately following π_{p+j} for some $0 \le j \le k$ where $p \in \mathrm{Pp}(\pi)$. Suppose that there are a starred plateau-peak blocks and b unstarred plateau-peak blocks to the right of π_{p+j}. According to the definition of the pp-maj-labeling of (π, S), we have $i = b(k+1) + j$. Inserting k copies of n at the position immediately following π_{p+j} increases $\text{pp-maj}(\pi)$ by $(k+1)(a+b)+j$. Moving the stars in a plateau-peak blocks one (plateau-peak) block to its left, the corresponding sum $\sum_{j\in \mathrm{Pp}(\pi)} (k+1)\cdot|S\cap\{1,2,\ldots,j-1\}|$ will be increased by $(k+1)a$. Hence, we have

$$\begin{aligned}
\text{pp-maj}(\sigma,S') &= \text{pp-maj}(\sigma) - |S'| - \sum_{j\in \mathrm{Pp}(\sigma)} (k+1)\cdot|S'\cap\{1,2,\ldots,j-1\}| \\
&= \text{pp-maj}(\pi,S) + (k+1)b + j \\
&= \text{pp-maj}(\pi,S) + i,
\end{aligned}$$

as desired.

Next, suppose that the position labeled i in the pp-maj-labeling of (π, S) is the position immediately following π_p for some $p \notin \overline{\mathrm{Pp}}(\pi)$. Suppose that there are a starred plateau-peak blocks and b unstarred plateau-peak blocks to the right of π_p, and c starred plateau-peak blocks and d unstarred plateau-peak blocks to the left of π_p. According to the definition of the pp-maj-labeling of (π, S), it is not difficult to check that $i = p - (k+1)c + (k+1)b$. Inserting k copies of n at the position immediately following π_p increases pp-maj(π) by $p + (k+1)(a+b)$ and increases the sum $\sum_{j \in \mathrm{Pp}(\pi)} (k+1) \cdot |S \cap \{1, 2, \ldots, j-1\}|$ by $(k+1)c$ as there are c starred plateau-peak blocks to the left of the first occurrence of n. Moving all the stars in a plateau-peak blocks one (plateau-peak) block to its left, the corresponding sum $\sum_{j \in \mathrm{Pp}(\pi)} (k+1) \cdot |S \cap \{1, 2, \ldots, j-1\}|$ will be increased by $(k+1)a$. Hence, we have

$$\begin{aligned} \text{pp-maj}(\sigma, S') &= \text{pp-maj}(\sigma) - |S'| - \sum_{j \in \mathrm{Pp}(\sigma)} (k+1) \cdot |S' \cap \{1, 2, \ldots, j-1\}| \\ &= \text{pp-maj}(\pi, S) + p - (k+1)c + (k+1)b \\ &= \text{pp-maj}(\pi, S) + i \end{aligned}$$

as desired.

In order to show that the map $\gamma_{n,k,s}$ is a bijection, we describe a map

$$\bar{\gamma}_{n,s,k} : \mathcal{XQ}_n^{\mathrm{Pp}}(k,s) \bigcup \mathcal{YQ}_n^{\mathrm{Pp}}(k,s) \longrightarrow \{0, 1, \ldots, kn - (k+1)s - k\} \times \mathcal{Q}_{n-1}^{\mathrm{Pp}}(k,s).$$

Let $(\sigma, S') \in \mathcal{XQ}_n^{\mathrm{Pp}}(k,s) \bigcup \mathcal{YQ}_n^{\mathrm{Pp}}(k,s)$.

Case (i′): If $(\sigma, S') \in \mathcal{YQ}_n^{\mathrm{Pp}}(k,s)$, we construct $\bar{\gamma}_{n,s,k}(\sigma, S') = (i, (\pi, S))$ by

- removing the stars in the rightmost plateau-peak block,
- removing all the n's,
- moving all the stars in each plateau-peak block one (plateau-peak) block to its right,
- labeling the positions in the leftmost plateau-peak block by stars and letting $i = (k+1)(r-s)$ where $r = \mathrm{pp}(\pi)$.

Case (ii′): If $(\sigma, S') \in \mathcal{XQ}_n^{\mathrm{Pp}}(k,s)$, we construct $\bar{\gamma}_{n,s,k}(\sigma, S') = (i, (\pi, S))$ by the following steps.

- Move all the stars in each plateau-peak block weakly to the right of the first occurrence of n (including the plateau-peak block containing n if n is a starred letter) one (plateau-peak) block to its right,

- Remove the n's and record i as the label of the position where $n's$ are inserted under the pp-maj-labeling of (π, S).

It is not difficult to check that cases (i$'$) and (ii$'$) in the construction of $\bar{\gamma}_{n,k,s}$ correspond, respectively, to cases (i) and (ii) in the construction of $\gamma_{n,k,s}$. Hence, the map $\gamma_{n,s,k}$ is indeed a bijection, completing the proof. ∎

Lemma 3.6 *For $n \geq 2$ and $s \geq 1$, there is a bijection*

$$\gamma^*_{n,k,s} : \{0, 1, \ldots, kn - (k+1)s\} \times \mathcal{Q}^{\mathrm{Pp}}_{n-1}(k, s-1) \longrightarrow \mathcal{Z}\mathcal{Q}^{\mathrm{Pp}}_{n}(k, s)$$

such that $\text{pp-maj}(\gamma^*_{n,k,s}(i, (\pi, S))) = \text{pp-maj}(\pi, S) + i$ *for any* $(\pi, S) \in \mathcal{Q}^{\mathrm{Pp}}_{n-1}(k, s-1)$ *and* $0 \leq i \leq kn - (k+1)s$.

Proof. First, we give a description of the map $\gamma^*_{n,k,s}$. Let $(i, (\pi, S)) \in \{0, 1, \ldots, kn - (k+1)s\} \times \mathcal{Q}^{\mathrm{Pp}}_{n-1}(k, s-1)$ with $\mathrm{pp}(\pi) = r$. In the following, we demonstrate how to generate $\gamma^*_{n,k,s}(i, (\pi, S))$.

Case (i): $i = (k+1)(r-s) - 1$ with $r \geq 1$ and the leftmost plateau-peak block is starred. We construct $(\sigma, S') = \gamma^*_{n,k,s}(i, (\pi, S))$ by the following procedure.

1. Remove the stars in the leftmost plateau-peak block;
2. Move all the stars in each plateau-peak block one (plateau-peak) block to its left;
3. Label the positions in the rightmost plateau-peak block by stars;
4. Let (π, T) be the resulting starred permutation;
5. Let (σ, S') be the starred permutation obtained from $\gamma_{n,k,s-1}(0, (\pi, T))$ by labeling the positions in its rightmost plateau-peak block by stars.

For example, if let $(\pi, S) = 12_*5_*5_*5_*24442_*3_*3_*3_*11$, then

$$(\pi, T) = 125552_*4_*4_*4_*2_*3_*3_*3_*11$$

and

$$\gamma^*_{6,3,3}(3, (\pi, S)) = 12_*6_*6_*6_*5552_*4_*4_*4_*2_*3_*3_*3_*11.$$

Case (ii): Otherwise, let $\gamma^*_{n,k,s}(i, (\pi, S))$ be a starred permutation obtained from $\gamma_{n,k,s-1}(i+1, (\pi, S))$ by labeling the positions in the rightmost plateau-peak block by stars. For example, if let $(\pi, S) = 12_*5_*5_*5_*24442_*3_*3_*3_*11$, then

$$\gamma^*_{6,3,3}(4, (\pi, S)) = 1_*6_*6_*6_*2_*5552_*4_*4_*4_*2_*3_*3_*3_*11.$$

By similar arguments as in the proof of the map $\delta^*_{n,k,s}$, one can verify that the map $\gamma^*_{n,k,s}$ is well-defined. Relying on Lemma 3.5, we can show that pp-maj$(\gamma^*_{n,k,s}(i,(\pi,S))) =$ pp-maj$(\pi,S)+i$ by the same reasoning as in the proof of Lemma 2.6.

In order to show the map $\gamma^*_{n,k,s}$ is a bijection, we give a description of a map $\bar{\gamma}^*_{n,k,s}$: $\mathcal{Z}\mathcal{Q}^{\mathrm{Pp}}_n(k,s) \longrightarrow \{0,1,\ldots,kn-(k+1)s\} \times \mathcal{Q}^{\mathrm{Pp}}_{n-1}(k,s-1)$. Let $(\sigma,S') \in \mathcal{Z}\mathcal{Q}^{\mathrm{Pp}}_n(k,s)$. In the following, we proceed to demonstrate how to generate $(i,(\pi,S)) = \bar{\gamma}^*_{n,k,s}(\sigma,S')$.

Let $(\tau,T) \in \mathcal{Q}^{\mathrm{Pp}}_{n-1}(k,s-1)$ be the unique starred permutation generated from (σ,S') by the following procedure.

- Remove all the stars in the rightmost plateau-peak block.
- Move all the stars in each plateau-peak block weakly to the right of the first occurrence of n (including the plateau-peak block containing n if n is a starred letter) one (plateau-peak) block to its right.
- Remove all the $n's$.

Record i' as the label of the position where $n's$ are inserted under the pp-maj-labeling of (τ,T).

Case (i′): If $i'=0$, then we generate $(i,(\pi,S))$ from (τ,T) by

- removing all the stars in the rightmost plateau-peak block,
- moving all the stars in each plateau-peak block one (plateau-peak) block to its right,
- labeling the positions in the leftmost plateau-peak block by stars and letting $i=(k+1)(r-s)-1$ where $r=\mathrm{pp}(\pi)$.

Case (ii′): Otherwise, set $(\pi,S)=(\tau,T)$ and $i=i'-1$.

It is not difficult to check that cases (i′) and (ii′) in the construction of $\bar{\gamma}^*_{n,k,s}$ correspond, respectively, to cases (i) and (ii) in the construction of $\gamma^*_{n,k,s}$. This implies that the maps $\gamma^*_{n,s,k}$ and $\bar{\gamma}^*_{n,s,k}$ are inverses of each other, and therefore the map $\gamma^*_{n,s,k}$ is indeed a bijection, completing the proof. ∎

The following result follows immediately from Lemmas 3.5 and 3.6.

Lemma 3.7 *For $n \geq 2$, the polynomial $\tilde{Q}^{\mathrm{Pp}}_{n,k,s}(q)$ equals 0 when $s<0$ or $s>n-1$ and*

$$\tilde{Q}^{\mathrm{Pp}}_{n,k,s}(q) = [kn-(k+1)s-k+1]_q \tilde{Q}^{\mathrm{Pp}}_{n-1,k,s}(q) + [kn-(k+1)s+1]_q \tilde{Q}^{\mathrm{Pp}}_{n-1,k,s-1}(q)$$

otherwise.

Now we are ready for the proof of Theorem 1.2.

Proof of Theorem 1.2. By Lemmas 3.3 and 3.7, we have $\tilde{Q}^{\mathrm{Pp}}_{n,k,s}(q) = Q^{\mathrm{Pp}}_{n,k,s}(q)$ since $\tilde{Q}^{\mathrm{Pp}}_{n,k,s}(q)$ satisfies the same recurrence and initial conditions as $Q^{\mathrm{Pp}}_{n,k,s}(q)$. Then Theorem 1.2 follows from Lemmas 3.1 and 3.4, completing the proof. ∎

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